

* Ideal diode on 3V3 to RP2040 does not work correctly when supply is less than 4V. Need a different solution here.
* Silkscreen for LED DAT/CLK is flipped
* Main board silkscreen for DAT[1-4] and CLK[1-4] give the wrong indication of the numbering... should be [4..1]

STM32_Micro
Microcontroller.SchDoc

DMX_RP2040
DMX_RP2040.SchDoc

DS
Drive Select.SchDoc

MC
Module Connector Female.SchDoc

SL
SmartLED.SchDoc

Low Power LED Module
LED_low_side_control.SchDoc

HP_LED
LED_high_power.SchDoc

AC/DC Power Supply

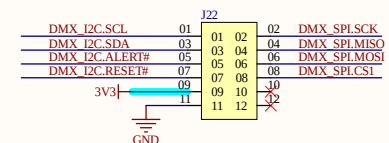
5V 200W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200A-5/7707187>

12V 162W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/LSP-160-12/11689253>

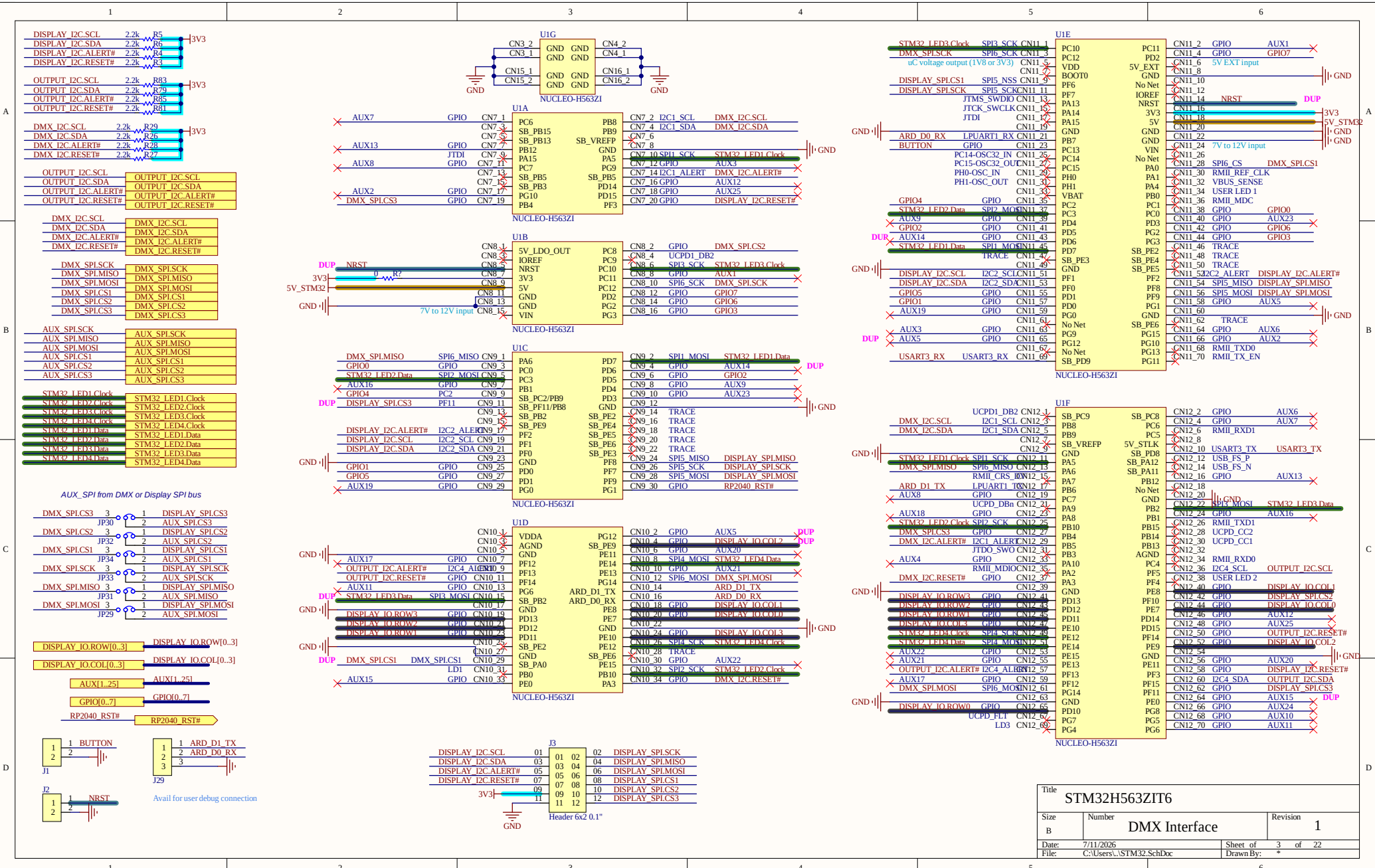
12V 200W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200-12/7707237>

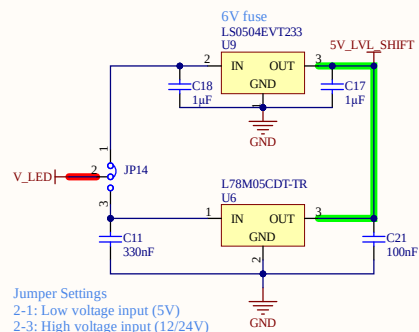
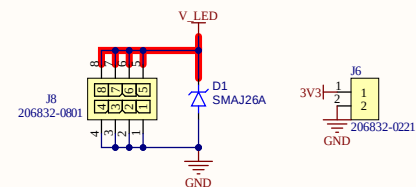
24V 162W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/LSP-160-24/11689249>

24V 200W (194.0mm x 55.0mm x 20.0mm)
<https://www.digikey.com/en/products/detail/mean-well-usa-inc/UHP-200-24/7707239>

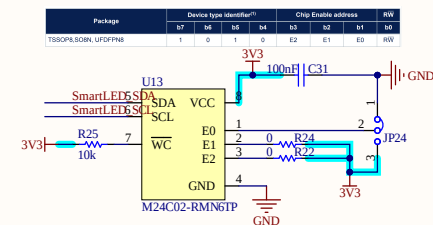
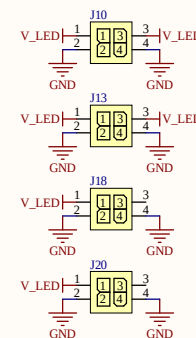
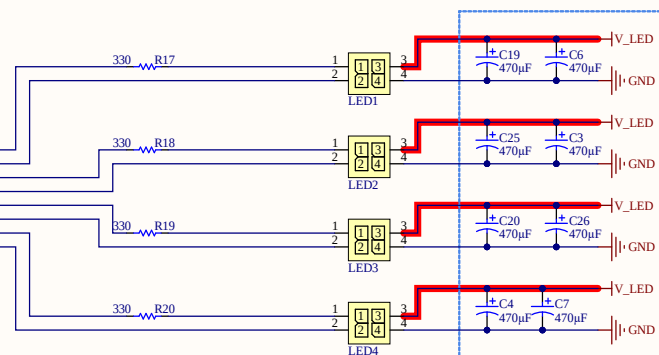
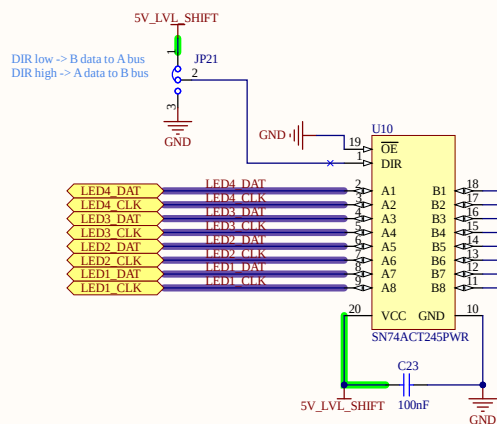
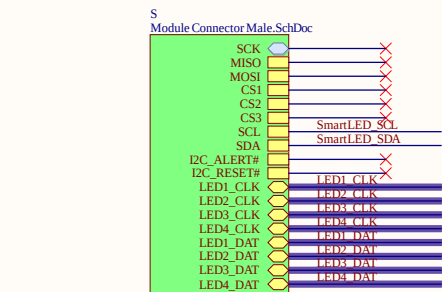


Title			
Block Diagram			
Size	Number	Revision	
B		DMX Interface	
Date:		7/11/2026	Sheet of 1 of 22
File:		C:\Users\...Block Diagram.SchDoc	Drawn By: *

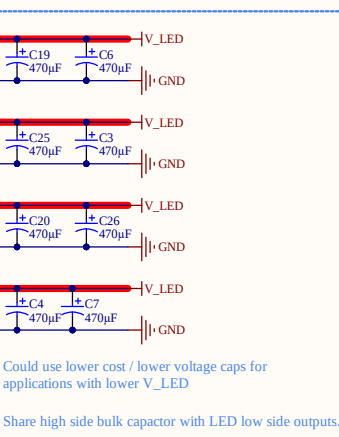




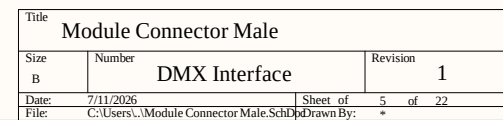
Alternate C93 cap 1uF 0603
CL10B105KB8NQNC

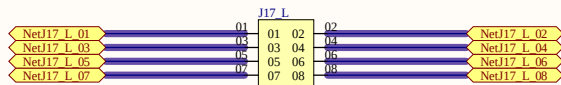
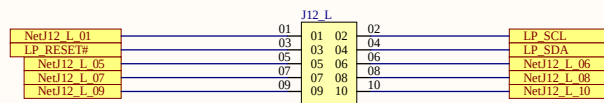
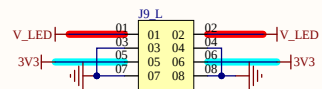


7-bit I2C address 0x50 through 0x57
Default to 0x56

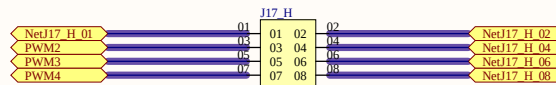
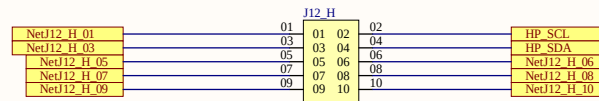
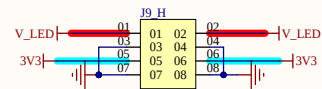


Title Smart LED			
Size B	Number DMX Interface		Revision 1
Date:	7/11/2026	Sheet of	4 of 22
File:	C:\Users\...\SmartLED.SchDoc	Drawn By:	*



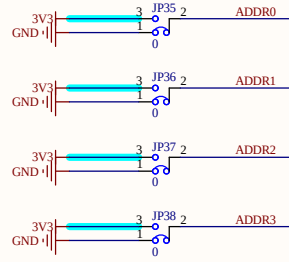
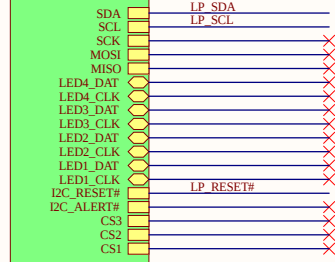


Title Module Connector Male			
Size	Number	Revision	
DMX Interface		1	
Date:	7/11/2026	Sheet of	5 of 22
File:	C:\Users\...\Module Connector Male.Sch Drawn By: *		

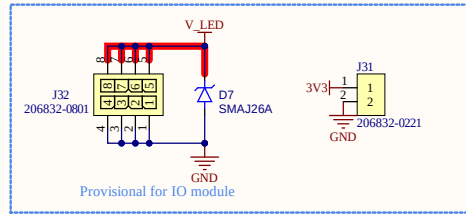


Title Module Connector Male				
Size B	Number DMX Interface			Revision 1
Date: 7/11/2026	Sheet of 5 of 22		File: C:\Users\...Module Connector Male.SchDoc	
Drawn By:				

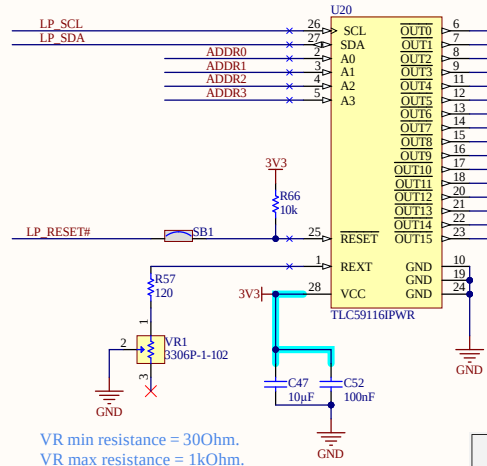
L Module Connector Male SchDoc



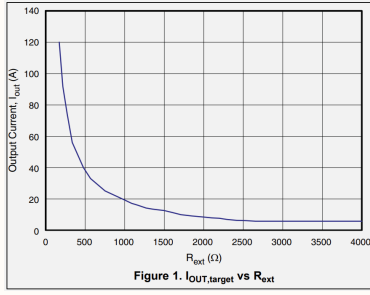
7-bit I2C address 0x60



Constant current LED sink driver
(12V 3W)
<https://www.ti.com/product/TLC59108>
<https://www.ti.com/product/TLC59116>

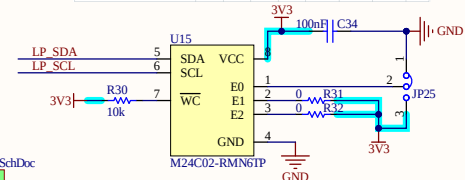


VR min resistance = 300ohm.
VR max resistance = 1kOhm.
Rext_min = 150 -> ~117mA
Rext_max = 1120 -> ~16mA



$$R_{ext} = 1.25V * 15 / I_{out}$$

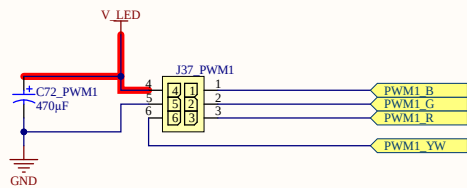
Package	Device type identifier ⁽¹⁾				Chip Enable address				SW
TSOP64/SO8N, UDFPN8	b7	b6	b5	b4	b3	b2	b1	b0	
	1	0	1	0	E2	E1	E0	000	



7-bit I2C address 0x50 through 0x57
Default to 0x56

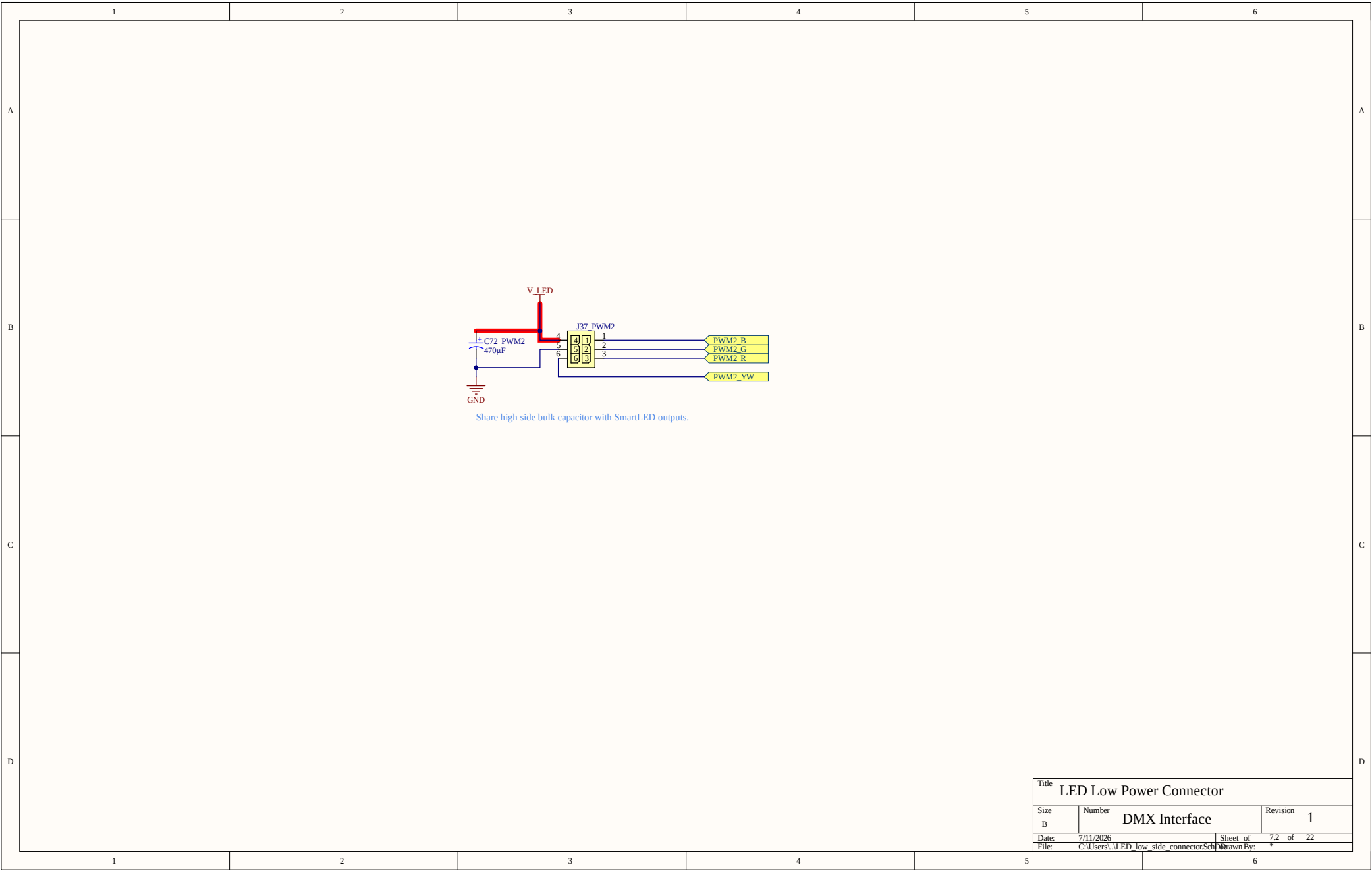


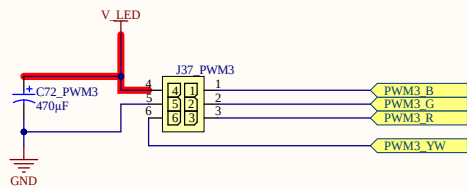
Title LED Low Power			
Size B	Number DMX Interface	Revision 1	
Date: 7/11/2026	Sheet of 6		of 22
File: C:\Users\LED_low_side_control\SchDoc		Drawn By:	



Share high side bulk capacitor with SmartLED outputs.

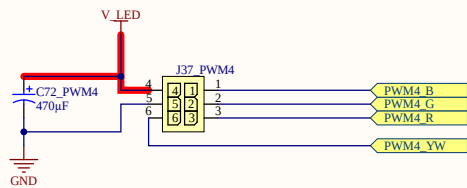
Title LED Low Power Connector		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	Sheet of 7.1 of 22	Drawn By: *
File: C:\Users\LED_low_side_connector\Sch		





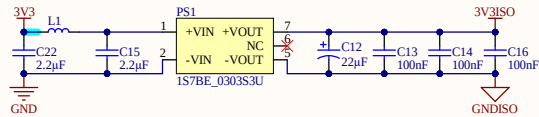
Share high side bulk capacitor with SmartLED outputs.

Title LED Low Power Connector		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	Sheet of 73 of 22	Drawn By:
File: C:\Users\LED_low_side_connector\Sch		

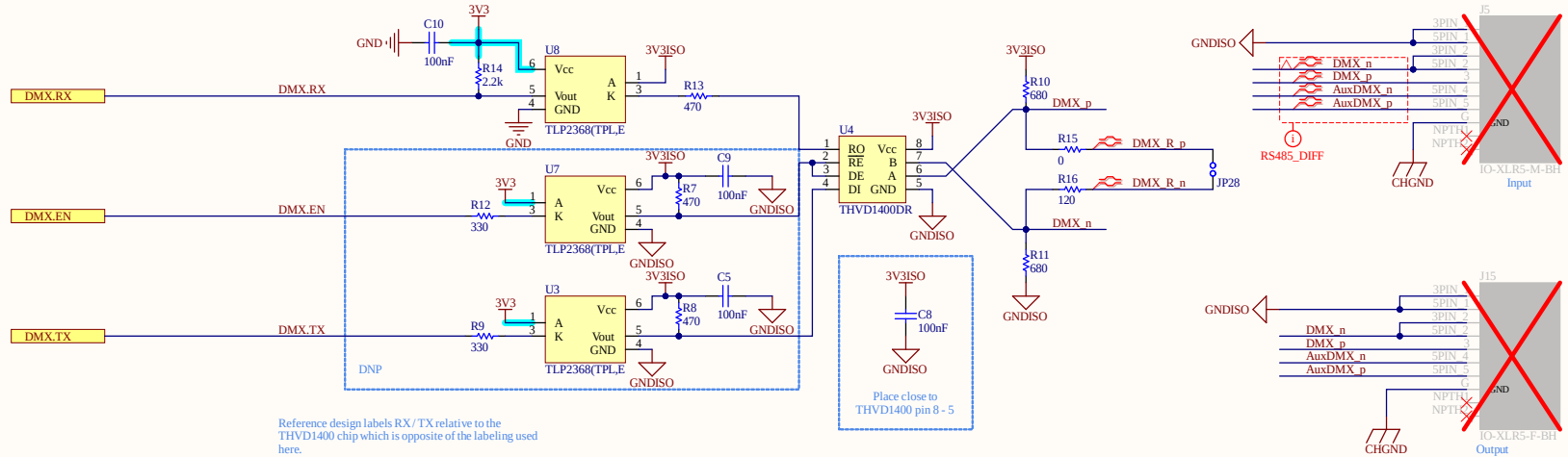


Share high side bulk capacitor with SmartLED outputs.

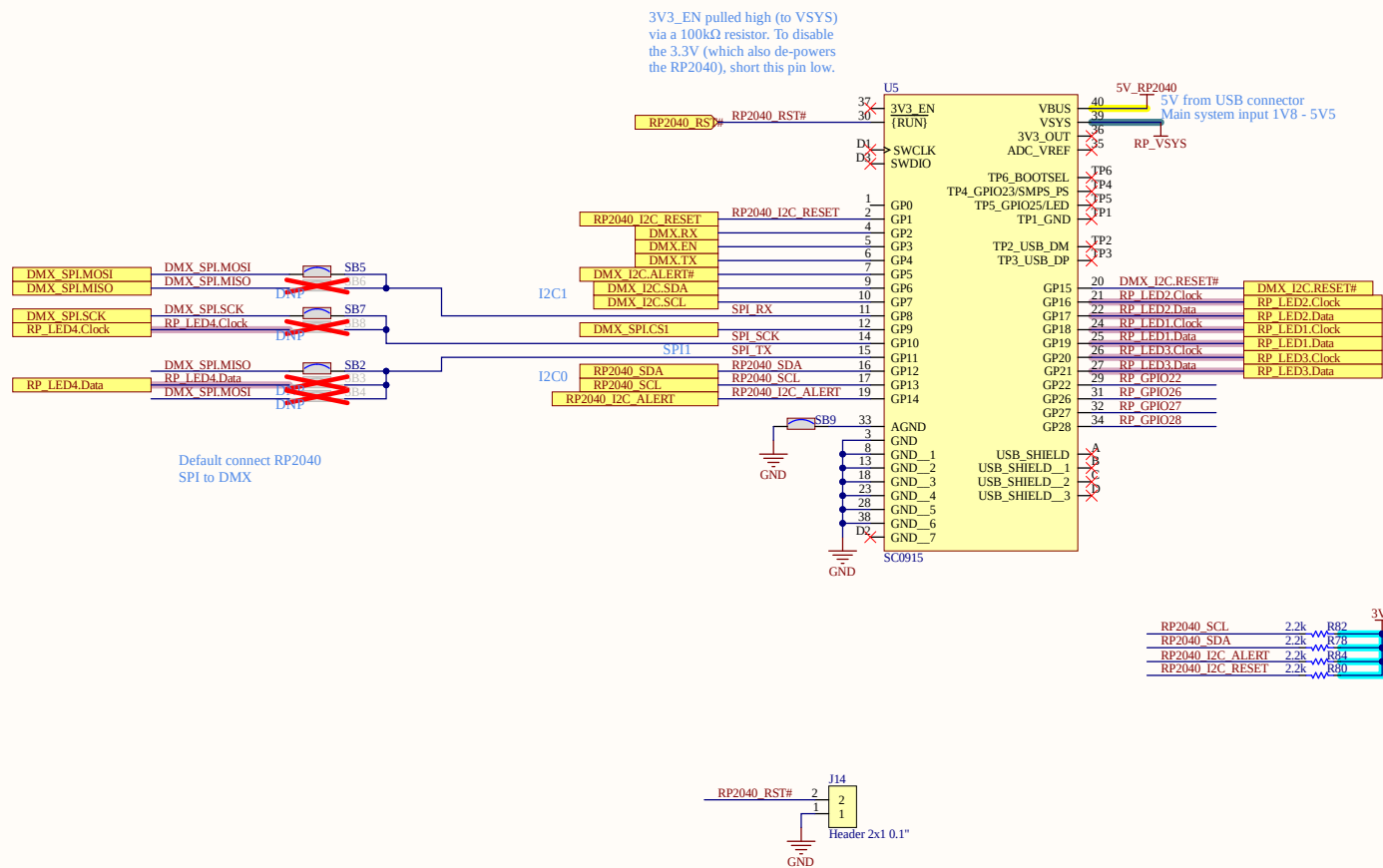
Title LED Low Power Connector		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	Sheet of 7.4 of 22	Drawn By:
File: C:\Users\LED_low_side_connector\Sch		



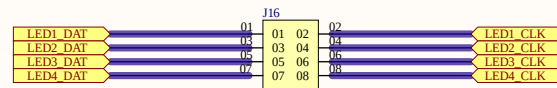
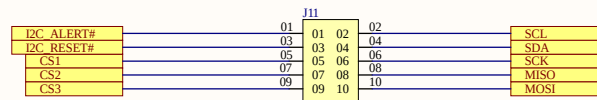
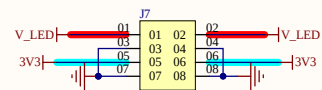
Connect 120 termination resistor. Place under output DMX connector.



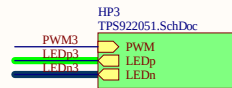
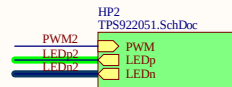
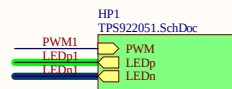
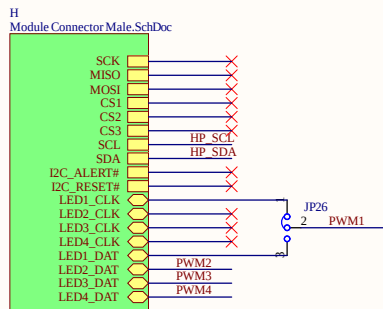
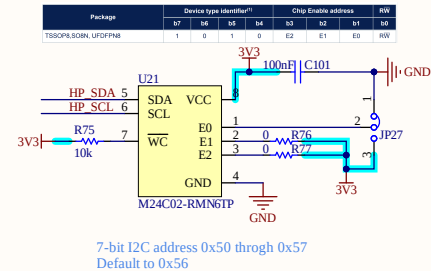
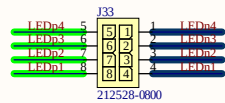
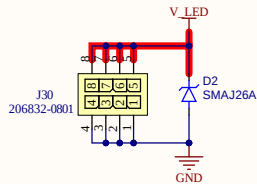
Title DMX / RS485			
Size B	Number DMX Interface	Revision 1	
Date: 7/11/2026	Sheet of 9 of 22		
File: C:\Users\...DMX.SchDoc	Drawn By:		



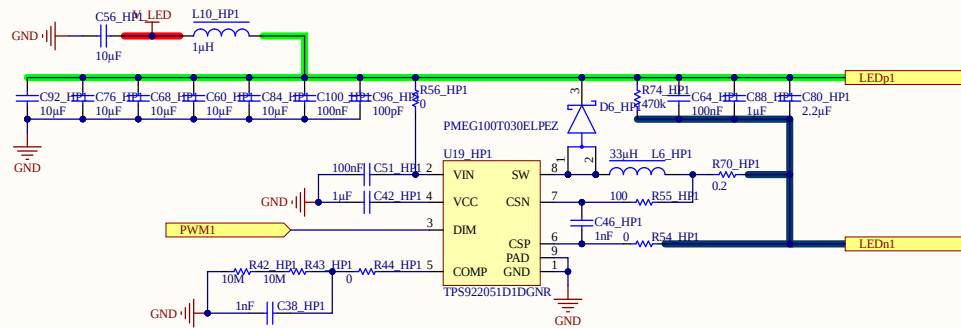
Title RP2040			
Size	Number	Revision	
B	DMX Interface	1	
Date:	7/11/2026	Sheet of	10 of 22
File:	C:\Users\...RP2040.SchDoc	Drawn By:	*



Title Module Connector Female			
Size	Number	Revision	1
DMX Interface			
Date:	7/11/2026	Sheet of	11 of 22
File:	C:\Users\...\Module Connector Female Sch	Drawn By:	*



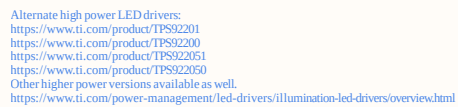
Title LED High Power		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	File: C:\Users\LED_high_power.SchDoc	Sheet of 12 of 22
		Drawn By: *



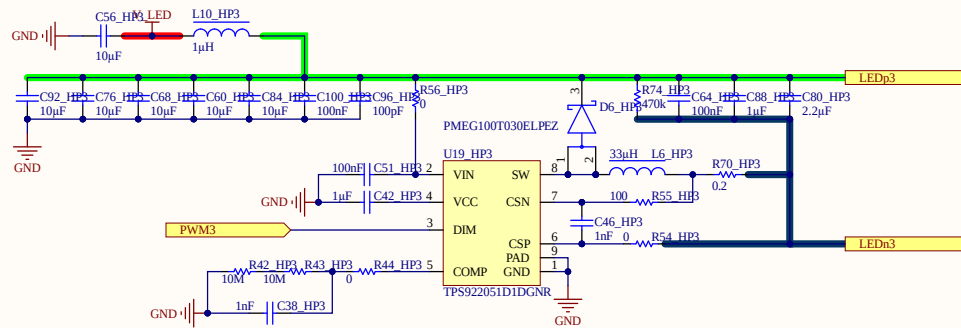
Alternate high power LED drivers:
<https://www.ti.com/product/TPS92201>
<https://www.ti.com/product/TPS92200>
<https://www.ti.com/product/TPS922051>
<https://www.ti.com/product/TPS922050>
 Other higher power versions available as well.
<https://www.ti.com/power-management/led-drivers/illumination-led-drivers/overview.html>

These are PWM dimmed.

Title LED High Power - TPS922051		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	Sheet of 13.1 of 22	Drawn By: *
File: C:\Users\... \TPS922051.SchDoc		



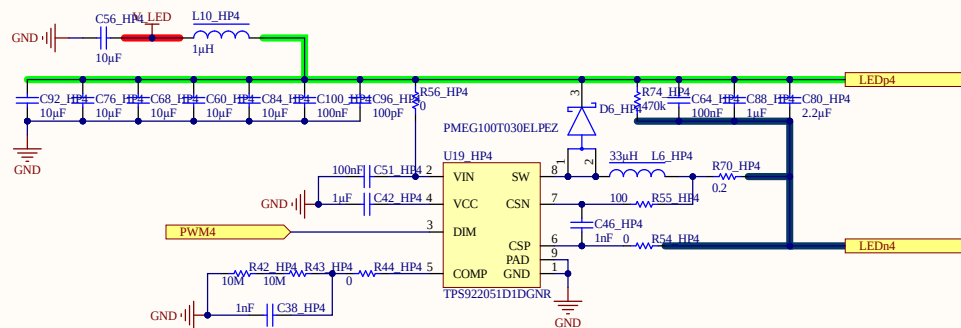
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Size		Number		Revision		1	
Date		7/11/2026		DMX Interface			
File:		C:\Users\...\TPS922051.SchDoc		Sheet of		13.2 of 22	
				Drawn By:		*	



Alternate high power LED drivers:
<https://www.ti.com/product/TPS92201>
<https://www.ti.com/product/TPS92200>
<https://www.ti.com/product/TPS922051>
<https://www.ti.com/product/TPS922050>
 Other higher power versions available as well.
<https://www.ti.com/power-management/led-drivers/illumination-led-drivers/overview.html>

These are PWM dimmed.

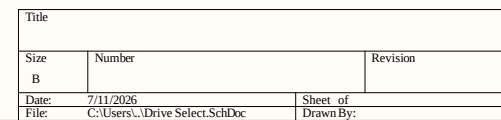
Title LED High Power - TPS922051		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	Sheet of 13.3 of 22	Drawn By: *
File: C:\Users\...\TPS922051.SchDoc		



Alternate high power LED drivers:
<https://www.ti.com/product/TPS92201>
<https://www.ti.com/product/TPS92200>
<https://www.ti.com/product/TPS922051>
<https://www.ti.com/product/TPS922050>
 Other higher power versions available as well.
<https://www.ti.com/power-management/led-drivers/illumination-led-drivers/overview.html>

These are PWM dimmed.

Title LED High Power - TPS922051		
Size B	Number DMX Interface	Revision 1
Date: 7/11/2026	Sheet of 13.4 of 22	Drawn By: *
File: C:\Users\... \TPS922051.SchDoc		



Board Stack Report